

An MLP-CycleGAN-based data-driven model for soft dielectric elastomer actuators with minimal Sim2Real gap

Xiazhi HU^{1,2}, Huangwei JI^{1,2}, Xiongfei FENG⁴, Jianing WU^{1,2}, Kai LUO^{1,2},
Xiangyang ZHU^{1,2,3} & Feifei CHEN^{1,2*}

¹ Robotics Institute, School of Mechanical Engineering, Shanghai Jiao Tong University, Shanghai 200240, China

² State Key Laboratory of Mechanical System and Vibration, Shanghai Jiao Tong University, Shanghai 200240, China

³ Meta Robotics Institute, Shanghai Jiao Tong University, Shanghai 200240, China

⁴ School of Materials Science and Engineering, Huazhong University of Science and Technology, Wuhan 430074, China

Received February 16, 2025; accepted April 9, 2025; published online June 17, 2025

Abstract Model-guided design of dielectric elastomer actuators (DEAs) is essential for enabling their application in soft robotics. However, current modeling methods primarily rely on the finite element method (FEM), which suffers from low computational efficiency. Additionally, the simulation-to-reality (Sim2Real) gap, mainly arising from variations in material properties and manufacturing processes, poses a significant challenge. In this work, we propose a data-driven modeling framework aimed at accurately and rapidly predicting voltage-induced displacements while minimizing the Sim2Real gap. The framework integrates a multi-layer perceptron (MLP) model, which serves as a computationally efficient surrogate for the FEM model, and a cycle-generative adversarial network (CycleGAN) model, which mitigates the Sim2Real gap by leveraging adversarial learning to process both simulation and experimental data. Dimensional analysis is performed to extend the framework's applicability across different DEA scales. The surrogate model delivers global predictions in just 0.8 s, achieving linear coefficients of determination (R^2) of 0.99106 for release distance prediction and 0.99375 for actuation distance prediction compared to experimental results. Our model can quickly identify the feasible range of biaxial prestretch ratios required for generating the desired deformation, thereby streamlining the design process. Finally, a soft robotic gripper is designed and fabricated, demonstrating versatile object-grasping capabilities.

Keywords soft robotics, dielectric elastomer actuators (DEAs), Sim2Real, CycleGAN

Citation: Hu X Z, Ji H W, Feng X F, et al. An MLP-CycleGAN-based data-driven model for soft dielectric elastomer actuators with minimal Sim2Real gap. *Sci China Tech Sci*, 2025, 68(7): 1720302, <https://doi.org/10.1007/s11431-025-2940-2>

1 Introduction

Soft robots can adapt to complex environments and perform various tasks due to their inherent compliance, deformability, and safety, often eliminating the need for complex sensors and control systems. Commonly used soft materials include dielectric elastomers (DEs), soft magnetic materials, liquid crystal elastomers, and shape memory alloys. Among them, DEs are particularly notable for their lightweight nature, high energy density, rapid response time, and high actuation frequency, making dielectric elastomer actuators (DEAs)

widely used as key components in various soft robotic systems for applications in grasping [1,2], swimming [3], and crawling locomotion [4,5]. A DEA typically consists of a DE membrane sandwiched between electrodes and supported by a compliant frame, operating under an applied voltage [6]. The voltage-induced deformation of DEAs primarily depends on DE's material properties [7], bi-axial prestretches [8], and the frame's structural characteristics [9]. However, the inherent material nonlinearity and electromechanical coupling effect of DEs, the complicated interaction between DE and the frame, as well as the geometric nonlinearity at large deformation, pose significant challenges in efficiently

* Corresponding author (email: fchen@sjtu.edu.cn)

and accurately designing DEAs to achieve the desired deformation performance [10].

Accurate and rapid prediction of deformation behavior is a critical prerequisite for effective DEA design [11]. For rigid robots and rigid-compliant hybrid robots, there are well-established modeling and analysis methods that can effectively predict the robot's motion, thereby guiding the design of robotic structures [12–15]. However, general analytical solutions are unavailable for most DEAs, necessitating the use of numerical approaches such as the finite element method (FEM). Previous studies have combined FEM with optimization algorithms to enhance the performance or achieve the desired motions of DEAs by adjusting the prestretching [16], spatial electric fields [17,18], frame configurations [19], or a combination of these factors [20]. These methods typically begin with a set of random initial solutions and iteratively update the parameters based on the gradient information. While this approach has proven effective for specific design tasks, it is time-consuming and limited to a case-by-case design paradigm, unable to generate a global design model. Additionally, the simulation-to-reality (Sim2Real) gap presents a significant challenge in DEA design [21]. This gap may arise from various factors, including material properties [22–24], manufacturing processes [25,26], assembly variations, environmental conditions [27], and time-dependent factors [28], which contribute to discrepancies between design predictions and actual performance. Therefore, developing an accurate and efficient deformation prediction model with a minimal Sim2Real gap is crucial to overcoming these challenges and advancing the design of DEAs.

Recently, machine learning (ML) approaches have shown great potential for constructing surrogate models of complex nonlinear systems [29]. They directly establish the relationship between inputs and outputs through data-driven methods, eliminating the need for physical models. Moreover, the trained surrogate models can achieve both high efficiency and high accuracy in predictions. For example, Li et al. [30] have used simulation data to build a surrogate model with a graph neural network (GNN) to efficiently and accurately predict the characteristic length and the actuation distance of DEAs with different frame parameters. Qin et al. [31] have proposed a deep learning-based design framework for DEAs that enables rapid and automatic generation of distributed electrode patterns for achieving the desired deformation. However, data-driven methods rely heavily on representative training data, and thus, surrogate models trained solely on simulation data still face the Sim2Real problem. The cycle-generative adversarial network (CycleGAN) has emerged as a promising solution to addressing the Sim2Real problem [32]. By utilizing adversarial learning, CycleGAN can establish a mapping between the simulation and experimental domains with a training dataset [33,34]. Moreover, in the context of imbalanced datasets, CycleGAN demonstrates

superior prediction accuracy and generalization capabilities compared to other machine learning methods [35,36].

Inspired by the achievements above, this study employs a data-driven approach to develop a fast and general global model for predicting the voltage-induced displacement field of DEA under varying prestretch ratios. To reduce the spatial dimension of design and network training costs, a dimensional analysis of the DEA is conducted. An FEM model is then built to generate a comprehensive simulation database, which is subsequently replaced by a multi-layer perceptron (MLP) surrogate model. The MLP model significantly accelerates predictions, reducing computation time from minutes with FEM to milliseconds. To address the Sim2Real problem, a CycleGAN model is trained using a small set of experimental data. This model accurately predicts the release and actuation states of DEAs under different prestretching conditions. The linear fit between the final surrogate model's predictions and the actual experimental results improved by 38.9% for the release distance and 725.3% for the actuation distance, compared with the FEM model. Furthermore, we showcase its application in grasping tasks, with the gripper's load capacity up to eight times its weight and a grasping range from paper to a nut approximately 10 mm wide. Overall, our method not only enhances prediction efficiency but also ensures the model's reliability in guiding experimental design, offering a promising solution for advancing DEA design and applications.

The rest of this article is described as follows. Section 2 introduces the basic structure, working principle, and dimensional analysis of DEA. Section 3 describes the FEM model, the MLP model, and their prediction results. Section 4 presents the experimental fabrication and measurement of the DEA, as well as the structure and prediction results of the CycleGAN model. Section 5 provides the application of DEA. Section 6 concludes the entire study.

2 Dimensional analysis of DEA

This section presents a dimensionless electromechanical analysis of the DEA to uncover its intrinsic complexities and identify key design parameters that are dimensionless and independent. This approach enhances the generality and applicability of the proposed data-driven prediction model.

A DEA typically consists of a prestretched DE membrane and a passive frame. In this work, we consider a common elliptical frame that can be morphed into a saddle-shaped structure by a prestretched DE membrane, for applications in soft grippers [1,2,37] and crawling locomotion [4,5]. The prestretching of the membrane and three states of DEA are shown in Figure 1. The x -axis and y -axis prestretch ratios of the DE membrane are denoted by λ_1^p and λ_2^p , respectively, as shown in Figure 1(a). The region inside the dashed line

represents the portion that will be bonded to the frame for making the actuator, while the area outside the dashed line will be removed. The semi-major axis and semi-minor axis lengths of the frame and membrane are denoted by a and b , respectively. After bonding the prestretched DE membrane to the frame, the overall structure of the DEA is shown in Figure 1(b). At this stage, the membrane remains in its initially prestretched state. Subsequently, the membrane is released, leading the DEA to morph into the released state as shown in Figure 1(c). The distance between the two ends of the actuator at this stage is defined as the release distance x_0 . The corresponding dimensionless parameter is $\tilde{x}_0$, where $\tilde{x}_0 = x_0 / a$. Under an applied voltage, the DEA reaches a new equilibrium state, referring to the actuated state shown in Figure 1(d), and the distance between the two ends of the actuator at this stage is defined as the actuation distance x_1 . The corresponding dimensionless parameter is $\tilde{x}_1$, where $\tilde{x}_1 = x_1 / a$. The distance increment, $x_1 - x_0$, represents the actuation stroke, for potential grasping and locomotion applications.

The equilibrium state of the DEA is determined by the interplay between the strain energies of the DE membrane and the frame. When DEA reaches equilibrium states, the total energy of the actuator is minimized. Thus the total variational energy is zero. The total variational energy can be divided into the variational energy of the DE membrane and the variational energy of the frame, denoted by e_m and e_f . The variational energy of the DE membrane e_m can be normalized at the prestretched domain Ω_m^p as follows:

$$e_m = \frac{\mu t_m a b}{\lambda_1^p \lambda_2^p} \int_{\Omega_m^p} \tilde{\mathbf{S}} \left(\mathbf{F}_1 \mathbf{F}_p, \frac{\varepsilon \tilde{E}^2}{\mu} \right) : \delta \mathbf{F}_1 \mathbf{F}_p d\tilde{\Omega}_m, \quad (1)$$

where μ and t_m denote the shear modulus and the initial thickness of the DE membrane, respectively; $\tilde{\mathbf{S}} = \mathbf{S} / \mu$ represents the dimensionless nominal stress;

$\mathbf{F}_p = \text{diag} \left(\lambda_1^p, \lambda_2^p, \frac{1}{\lambda_1^p \lambda_2^p} \right)$ is the deformation gradient from the initial state to the prestretched state, $\mathbf{F}_1$ is the deformation gradient from the prestretched state to the released state.

Next, since the passive frame is subjected to in-plane bending, it is modeled using the Euler-Bernoulli beam theory. The variational energy of the passive frame e_f can be expressed by

$$e_f = \frac{E_f w_1 t_f^3}{12 C_0} \int_{\tilde{C}} \tilde{\kappa} \delta \tilde{\kappa} d\tilde{s}, \quad (2)$$

where E_f , w_1 , and t_f are the Young's modulus, the width, and thickness of the passive frame, respectively; C_0 is the perimeter of the ellipse, which is proportional to a ; $\tilde{C}$ represents the normalized center line of the passive frame; $\tilde{\kappa}$ represents

the dimensionless curvature, where $\tilde{\kappa} = \kappa C_0$; $\tilde{s}$ represents the normalized arc length of the passive frame, where $\tilde{s} = s / C_0$.

To simplify the calculations, we define $e = b / a$, $k_1 = w_1 / a$, $k_2 = w_2 / a$. Then, by calculating the equilibrium equations, the critical dimensionless parameters of the DEA are obtained. The dimensionless parameters and their physical significances are presented in Table 1. We choose $\frac{t_m a^2}{t_f^3}$,

λ_1^p , and λ_2^p as design variables. The influence of the geometric parameters of the frame and membrane on the equilibrium state of the DEA is captured by the first dimensionless parameter $\frac{t_m a^2}{t_f^3}$, along with the fixed parameters e , k_1 , and k_2 . The two dimensionless parameters λ_1^p

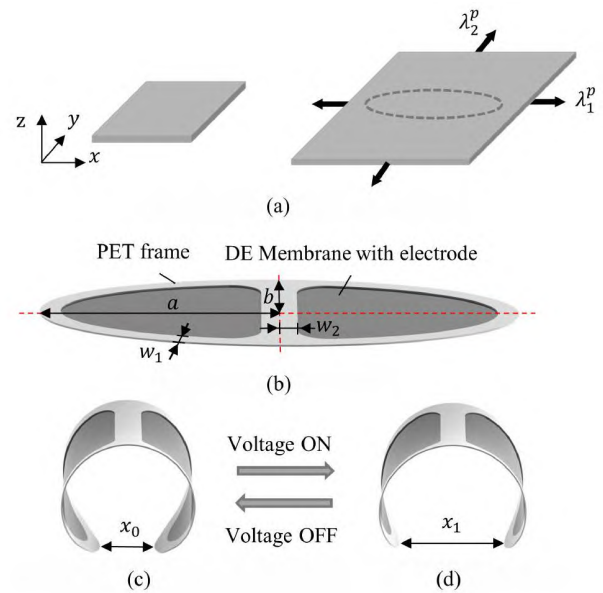


Figure 1 (Color online) Prestretched, released, and actuated states of DEA. (a) The DE membrane at the initial unstretched state (left) and the state prestretched by λ_1^p and λ_2^p along the x -axis and y -axis (right). (b) The prestretched DE membrane is bonded to the elliptical frame. (c) The released state of DEA, with the release distance x_0 . (d) The actuated state of DEA, with the actuation distance x_1 .

Table 1 Dimensionless quantities of a DEA module

Parameter	Physical significance
$\frac{t_m a^2}{t_f^3}$	Geometry of the actuator
λ_1^p	x -axis prestretch ratio of DE
λ_2^p	y -axis prestretch ratio of DE
$\frac{\mu}{E_f}$	Material modulus ratio
$\frac{\varepsilon \tilde{E}^2}{\mu}$	The dimensionless Maxwell stress

and λ_2^p reflect the effect of adjusting the membrane's pre-stretches on the equilibrium state of the DEA, which are the main design variables investigated in this work. $\frac{\mu}{E_f}$ represents the material modulus ratio, and $\frac{\varepsilon \bar{E}^2}{\mu}$ represents the dimensionless Maxwell stress, where ε and $\bar{E}$ denote the absolute permittivity and the electric field strength. These two variables remain unchanged in the design process.

3 Surrogate model of DEA by MLP

In this section, we first construct the FEM model of the DEA. Then, based on the simulation results, the value of the dimensionless design variable $\frac{t_m a^2}{t_f^3}$ is determined. Utilizing the determined value of $\frac{t_m a^2}{t_f^3}$ can facilitate the extension of feasible design parameter schemes for the target configuration. Subsequently, parameter sweeps are performed on the two dimensionless design variables, λ_1^p and λ_2^p , to generate a comprehensive simulation dataset. Finally, an MLP model is constructed, with the simulation data used for training and testing. The MLP model's predictions closely match the simulation results, with a significant improvement in prediction speed.

3.1 FEM model

The FEM simulation is performed in the commercial software ABAQUS. To improve computational efficiency, only one-quarter of the DEA structure is analyzed due to its symmetry. The FEM model of DEA includes a membrane and a frame, with the membrane positioned on the same side of the frame. Membrane elements and shell elements are used to represent the membranes and frame, respectively, with tie constraints applied to their interactions. The Gent model is employed to describe the membrane's material properties, and the user subroutine UMAT is employed for capturing the stress-strain relationship of the DE material [8,16]. The material parameters used in the simulation are provided in Table 2. The simulation involves three analysis steps: prestretching, release, and actuation, corresponding to the three states of the actuator illustrated in Figure 1.

After completing the construction of the FEM model, the dimensionless design variable $\frac{t_m a^2}{t_f^3}$ is first determined. According to the previous dimensional analysis, $\frac{t_m a^2}{t_f^3}$ is an in-

dependent variable that influences the equilibrium configuration of the DEA. Considering the target configuration of DEA and the feasibility of experimental fabrication, the determined geometric parameters of the DEA are provided in Table 3, corresponding to a specific value of $\frac{t_m a^2}{t_f^3}$. Notably, this dimensionless parameter guides the design of DEAs across various sizes. Specifically, by adjusting the values of t_m , a , and t_f , while keeping this dimensionless parameter constant, the corresponding dimensionless distance of the DEA remains unchanged. This approach extends the dimensionless distance results from a single case to various sizes, thus expanding the feasible design space with minimal effort.

The two key design variables are λ_1^p and λ_2^p . Considering that large prestretch values often cause localized stress concentrations that result in simulation failures, the traversal ranges for λ_1^p and λ_2^p are selected by balancing the simulation success rate and the feasibility to yield the target configuration, as shown in Table 3. The traversal is performed in ABAQUS with a step size of 0.05 for both λ_1^p and λ_2^p , and λ_1^p is set to be greater than λ_2^p for generating the target configuration. We observe that configurations with λ_1^p less than λ_2^p result in almost planar structures after the release step and exhibit a small actuation stroke, which may be due to the dominant bending deformation occurring in the yo z plane rather than the xo z plane. A total of 352 valid simulation results are obtained, as shown in Figure 2(a) and (b). The entire analysis required approximately 8 h to complete.

Table 2 Material parameter settings of the DEA

Parameter	Value
Thickness of the passive frame t_f (mm)	0.35
Young's modulus of the passive frame E_f (MPa)	2300
Initial thickness of the DE membrane t_m (mm)	0.1
Shear modulus of the DE membrane μ (MPa)	0.32
Stretch limit of the DE membrane J_m	10
Permittivity of the DE membrane ε (F m ⁻¹)	2.8×10^{-11}

Table 3 The determined parameters of the DEA

Parameter	Value
e	0.5
k_1	0.0875
k_2	0.075
a	40
λ_1^p	[1.1,2.5]
λ_2^p	[1.1,2.2]

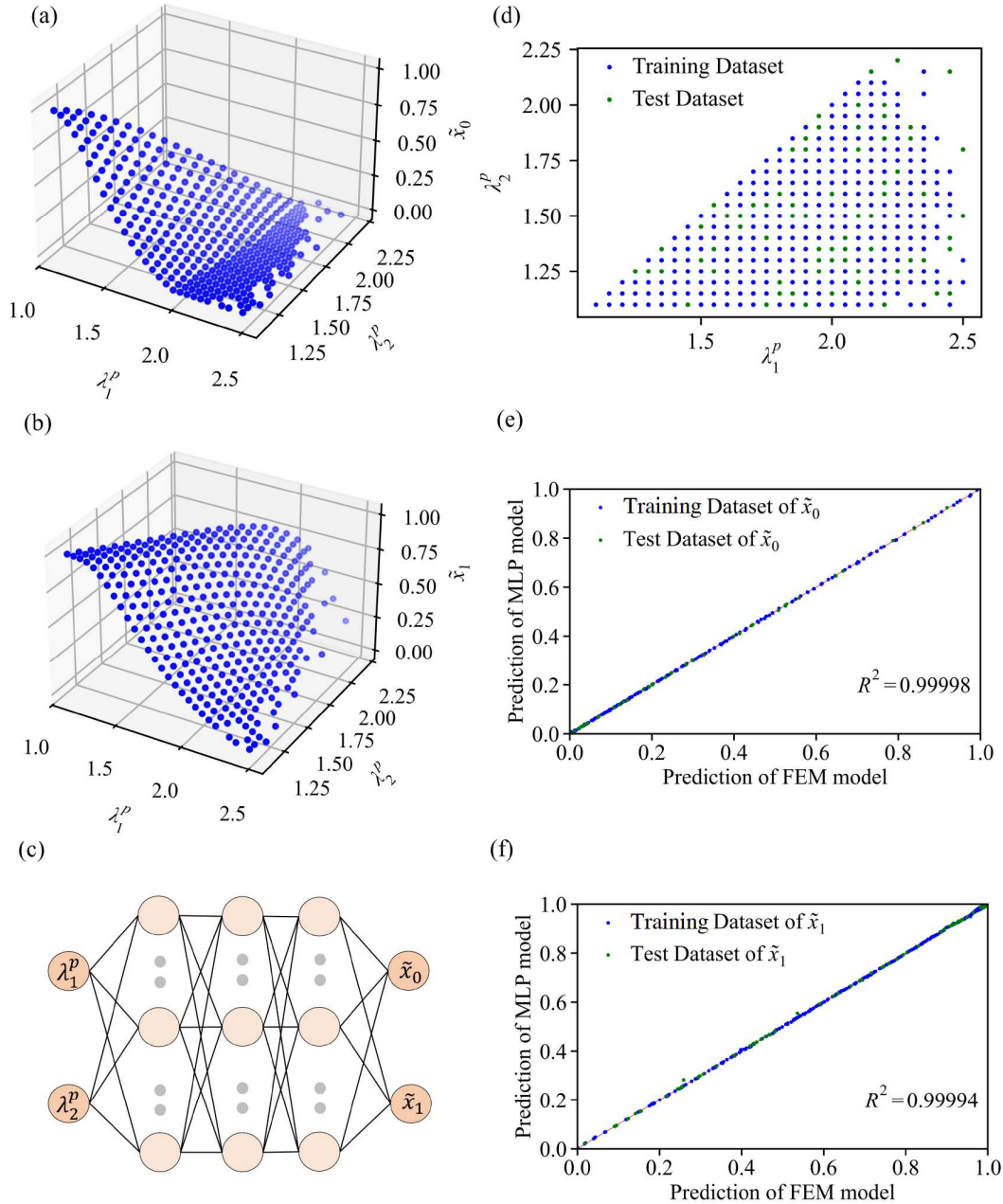


Figure 2 (Color online) Surrogate model of DEA by MLP. (a) Predicted release distance from the FEM model; (b) predicted actuation distance from the FEM model; (c) structure of the MLP model; (d) distribution of sample design parameters for the MLP model; (e) comparison of the MLP model's predicted release distance with simulation results from the FEM model; (f) comparison of the MLP model's predicted actuation distance with all simulation results from the FEM model.

3.2 MLP model

The basic structure of the MLP model consists of an input layer, hidden layers, and an output layer. Both the input and output layers contain two neurons, representing the input bi-axial prestretch ratios λ_1^p , λ_2^p and the output distance parameters x_0 , x_1 , respectively. To balance the prediction accuracy and computational efficiency, the hidden layers are structured with three layers containing 20, 16, and 10 neu-

rons, respectively. The designed MLP structure is shown in Figure 2(c). Linear connections are used between the input layer and the hidden layers, as well as between the hidden layers and the output layer. Sigmoid activation functions are applied between the hidden layers. The MLP model is implemented in Python using the PyTorch framework.

For training, the 352 simulation datasets are randomly divided into a training dataset (80%) and a test dataset (20%), as shown in Figure 2(d). The L2 loss function is adopted to

measure the loss, and the Adam optimizer is employed for parameter optimization. Training the MLP model takes approximately 14 s. For verification, the linear coefficient of determination R^2 is used to evaluate the linear fit between the MLP model's predictions and the FEM model's predictions. R^2 closer to 1 reflects a stronger linear correlation, meaning the MLP model's predictions closely match the FEM model's predictions. The fitting results between the MLP model and FEM model are shown in Figure 2(e) and (f), with R^2 values of 0.99998 and 0.99994 for the release distance and actuation distance, respectively. Each point in Figure 2(e) and (f) represents a dataset corresponding to specific prestretch ratios. If a point is close to the red reference line with a slope of 1, it indicates that the predictions made by the MLP surrogate model for that point closely match the simulation results. Moreover, the trained MLP model predicts all 352 cases within 0.02 s. These results demonstrate that the MLP model, as a surrogate model for the FEM model, significantly accelerates the prediction speed while maintaining high accuracy for displacement predictions across various prestretch ratios in the simulation environment.

4 Minimizing Sim2Real gap by CycleGAN model

4.1 Fabrication and experimental characterization

Using the prestretch ratio as the variable, 25 representative cases are selected, and the corresponding DEAs are fabricated accordingly. To achieve a relatively high actuation force and stability, ELASTOSIL® Film 2030 is chosen as the DE membrane material, while polyethylene terephthalate (PET) is selected for the frame. Additionally, each DEA comprises two membrane layers and one PET layer. The schematic of the DEA fabrication process is shown in Figure 3(a). The PET frame is cut using a laser cutting machine (VLS3.50-SYS) according to the shape and dimensions specified in Table 3. The membrane preparation begins with precise prestretching of the DE membrane using a customized stretching machine [16]. The prestretched membranes are then transferred from the stretching machine onto acrylic frames, resulting in two separate prestretched DE membranes. Lastly, carbon grease is applied to both the upper and lower surfaces of each membrane. Before adhesion, conductive tapes are first applied to the conductive regions of each membrane. The membranes are then bonded together by aligning the conductive regions. Finally, the frame is attached. The adhesive used is ELASTOSIL® E41, with a curing time of two hours. After the adhesive cures, the excess membrane around the passive frame and acrylic frame is removed, and the DEA is fixed to a support frame, as shown in Figure 3(a).

We then conduct experimental characterization of the DEA's voltage-induced displacement. The displacement at its tip follows a curved trajectory and varies depending on the prestretch ratios. A photographic measurement method is used, as shown in Figure 3(b) and (c), for capturing the tip positions at the initial, released, and actuated states. The high-voltage power supply is controlled by a dSPACE-DS1103 control board equipped with digital-to-analog converters. The output voltage follows a ramp function with an adjustable slope and maximum value. When recording the experimental data, the slope is set to 600 V s^{-1} , with a maximum value of 6000 V. The comparison between the MLP model's predicted results and the experimental data is shown in Figure 3(d) and (e). It is evident that there is a significant error between them, indicating that the MLP model cannot directly guide experimental design.

4.2 CycleGAN model

The CycleGAN model typically processes data from two domains (e.g., the Sim domain and the Real domain) and establishes a mapping between them, making this method particularly suitable for addressing the Sim2Real problem. The workflow of the entire surrogate model is shown in Figure 4(a), where the dashed blue box illustrates the detailed training process of the CycleGAN model. The CycleGAN model consists of generators (G_{S2R} and G_{R2S}) and discriminators (D_S and D_R).

The network structure of the generators is shown in Figure 4(b), where the inputs are prestretch ratios, and the outputs are release distance and actuation distance. The discriminator's network structure is depicted in Figure 4(c), with prestretch ratios and corresponding distance results as the input, and the probability of authenticity as the output. The goal of the generators is to produce data that closely resemble the target domain, while the discriminators aim to differentiate between generated and original data. For example, D_S differentiates between simulation data generated by G_{R2S} and simulation data predicted by the MLP model, with the former ideally yielding a result close to 0 and the latter close to 1. Meanwhile, D_R functions similarly. During training, the generators and discriminators optimize their parameters through adversarial competition using the backpropagation algorithm, ultimately yielding a generator that accurately maps all Sim domain data to the Real domain.

The training process employs three types of loss functions: adversarial loss, cycle consistency loss, and targeted regression loss. The adversarial loss is used to primarily evaluate the discriminator's ability to distinguish data, indirectly reflecting the generator's performance. The adversarial loss [38] we adopted is expressed as follows:

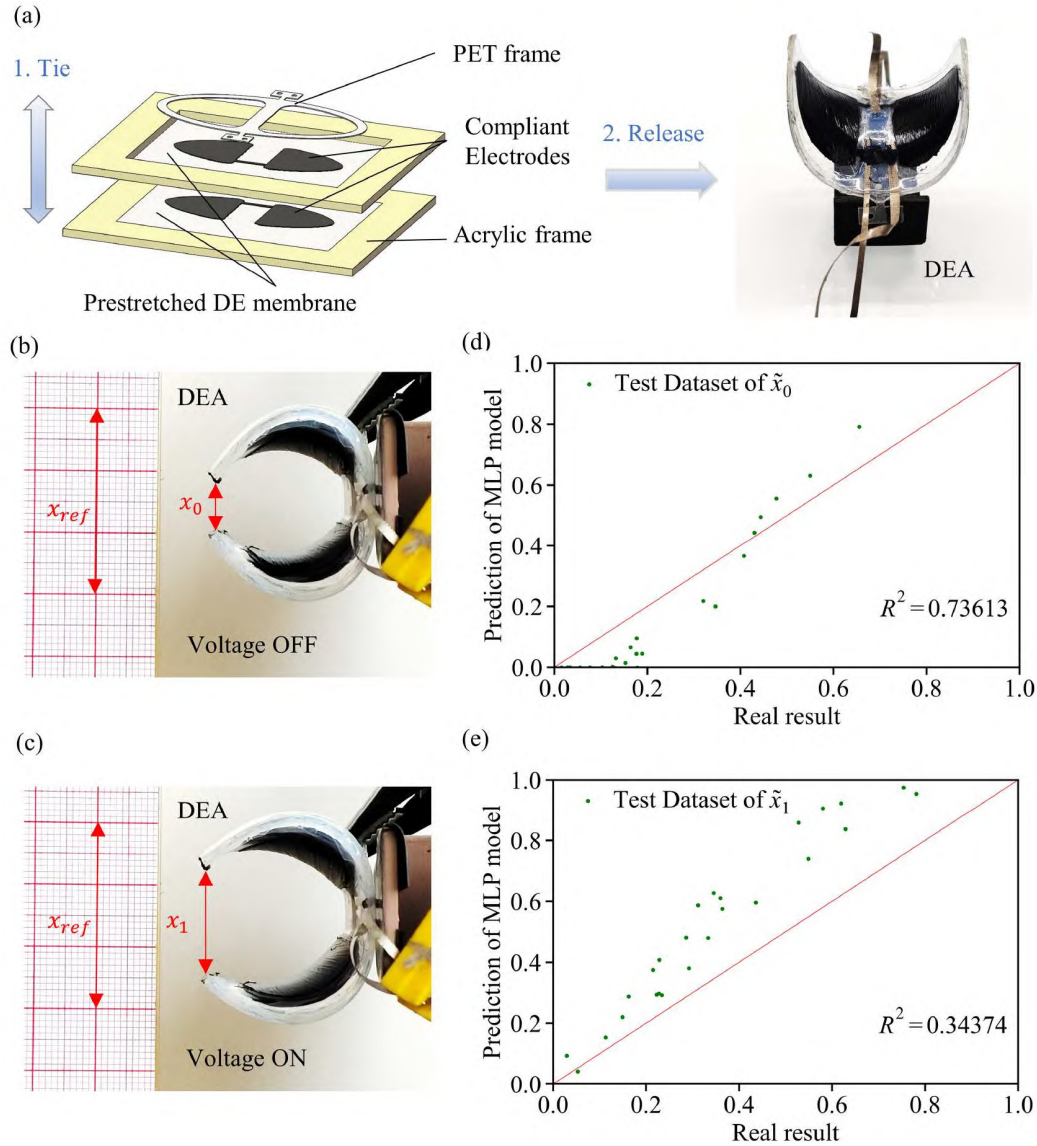


Figure 3 (Color online) Fabrication and experimental characterization of the DEA. (a) Components of the DEA assembly (left), with the acrylic frame serving as prestretch support, and its released configuration (right). The distance between the two tips of the DEA in the released state (b) and actuated state (c) is measured by comparing it with the reference distance, $x_{ref} = 30$ mm. The comparison between the MLP predicted results and the experimental data for (d) the release distance and (e) the actuation distance.

$$L_{GAN(S2R)}(G_{S2R}, D_R) = E_{r \sim P_{data}(R)} \left[(1 - D_R(r))^2 \right] + E_{s \sim P_{data}(S)} \left[(D_R(G_{S2R}(s)))^2 \right], \quad (3)$$

$$L_{GAN(R2S)}(G_{R2S}, D_S) = E_{s \sim P_{data}(S)} \left[(1 - D_S(s))^2 \right] + E_{r \sim P_{data}(R)} \left[(D_S(G_{R2S}(r)))^2 \right], \quad (4)$$

with s the simulation data sampled from $P_{data}(S)$ and r the real-world data sampled from $P_{data}(R)$.

The cycle consistency loss is used to constrain the possible mapping relationships established by the generator between

the two domains, preventing many-to-one mappings. It ensures that any data s in the Sim domain should remain the same after two mappings through G_{S2R} and G_{R2S} , and similarly for the Real domain. The cycle consistency loss [33] we adopted is expressed as follows:

$$L_{cyc}(G_{S2R}, G_{R2S}) = E_{s \sim P_{data}(S)} \left[\|G_{R2S}(G_{S2R}(s) - s)\|_2 \right] + E_{r \sim P_{data}(R)} \left[\|G_{S2R}(G_{R2S}(r) - r)\|_2 \right]. \quad (5)$$

The targeted regression loss is used to ensure that the generator establishes mappings based on paired simulation and real data, avoiding unpaired mappings. The targeted regression loss [35] used is as follows:

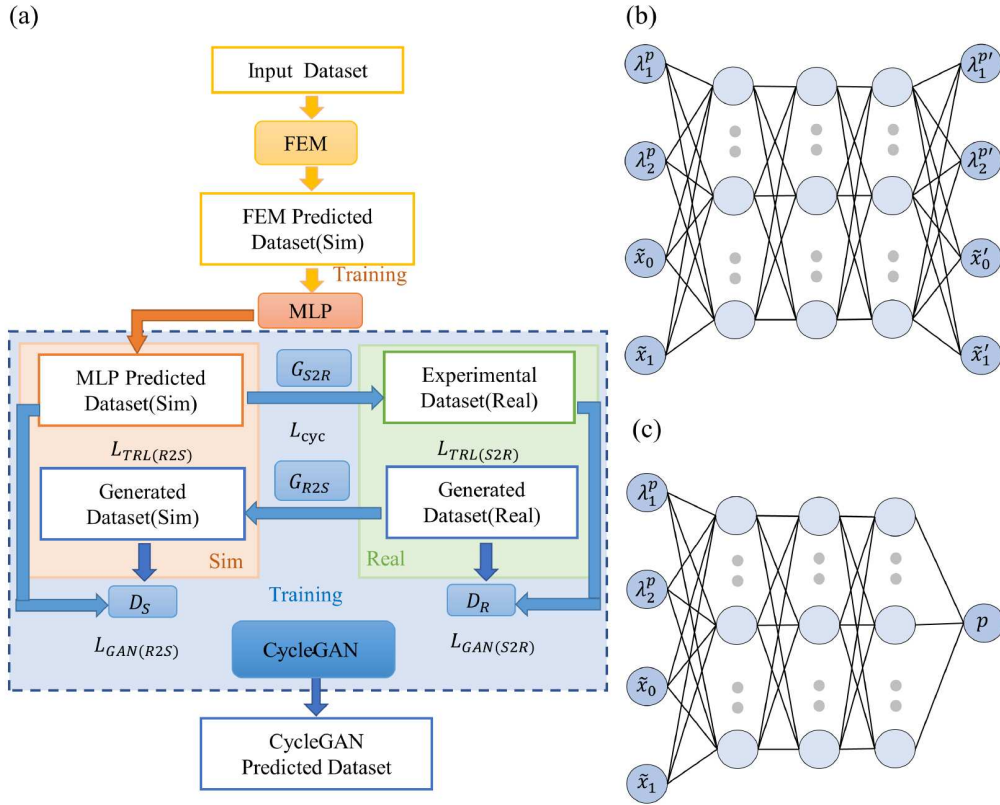


Figure 4 (Color online) The surrogate model capable of accurately predicting real-world data. (a) The workflow of the surrogate model, with the dashed blue box detailing the training process of the CycleGAN model. The detailed network structures of the generators G_{S2R} and G_{R2S} in the CycleGAN model are shown in (b), while the detailed network structures of the discriminators D_S and D_R are presented in (c).

$$L_{TRL(S2R)} = \frac{1}{n} \sum_{i=1}^n \|G_{S2R}(s_i) - r_i\|_2, \quad (6)$$

$$L_{TRL(R2S)} = \frac{1}{n} \sum_{i=1}^n \|G_{R2S}(r_i) - s_i\|_2. \quad (7)$$

From the 25 experimental data, 8 representative data are chosen as the training dataset, with the rest used for testing, as shown in Figure 5(a). The training process, using the Adam optimizer, consists of three steps. In the first step, experimental data are excluded, and only simulation data are utilized in both domains, aiming to learn the self-mapping relationship of the Sim domain. This approach helps reduce the generator's possible mapping relationships, thereby improving the prediction accuracy. The total loss function used here involves only L_{GAN} and L_{cyc} . In the second step, all simulation and experimental data are introduced without considering pairing, so the total loss function still involves only L_{GAN} and L_{cyc} . In the third step, L_{TRL} is added to the total loss function, and only the eight experimental training datasets and their corresponding simulation data are used. Training the CycleGAN model takes about 15 s. The final CycleGAN model predicts 352 data points in 0.8 s. The fitting results between the CycleGAN model predictions and

experimental data are shown in Figure 5(b) and (c), with R^2 values of 0.99106 and 0.99375, respectively, indicating a significant improvement in prediction accuracy by incorporating the CycleGAN model into the surrogate model. A comparison of predicted results of FEM, MLP, and CycleGAN is listed in Table 4.

4.3 Validation of surrogate model

As shown in Figure 6, our proposed model, integrating surrogate modeling and Sim2Real correction, enables global prediction of DEA states (including released and actuated states) under various prestretch conditions. Using this model, feasible prestretch parameters can be efficiently selected to meet specific displacement requirements. For instance, the red points in Figure 6 identify feasible prestretch ratios for achieving an actuation distance greater than 15 mm and a release distance smaller than 15 mm, with their difference representing the operational range. Within the feasible solution space, the operational range of the gripper can be directly determined, facilitating parameter selection for the design process. For example, among the feasible solutions, feasible solution 1, with prestretch ratios of $\lambda_1^p = 2.2$ and

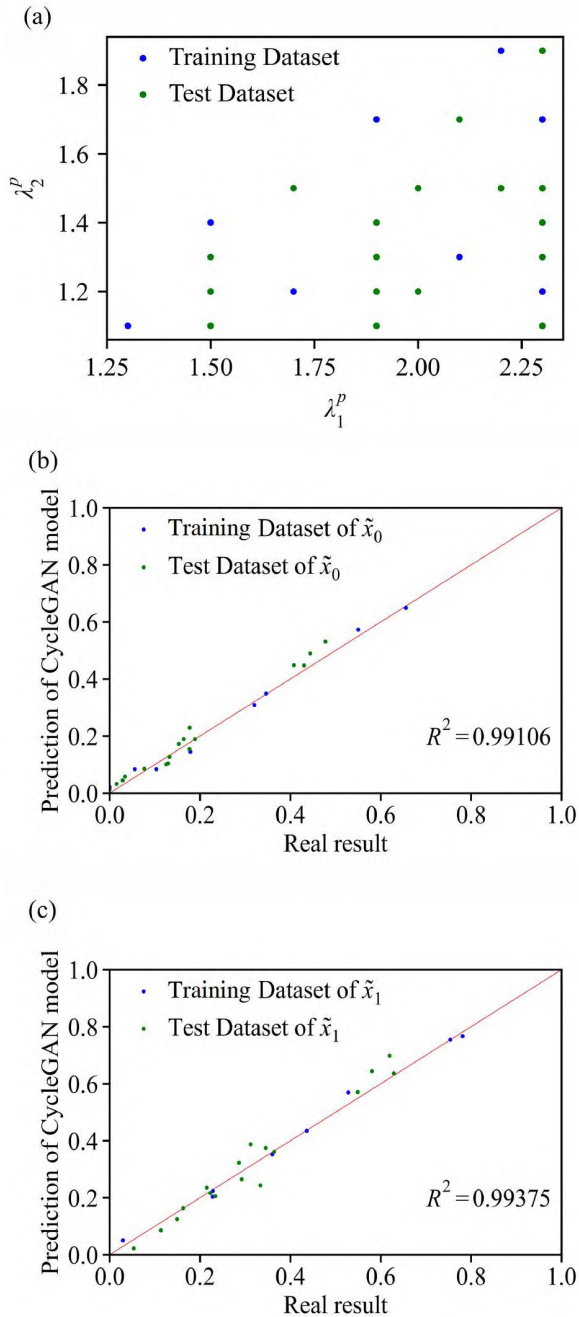


Figure 5 (Color online) Dataset division and testing of the CycleGAN model. (a) Division and distribution of sample design parameters; (b) fitting result of release distance between CycleGAN model and experiment; (c) fitting result of actuation distance between CycleGAN model and experiment.

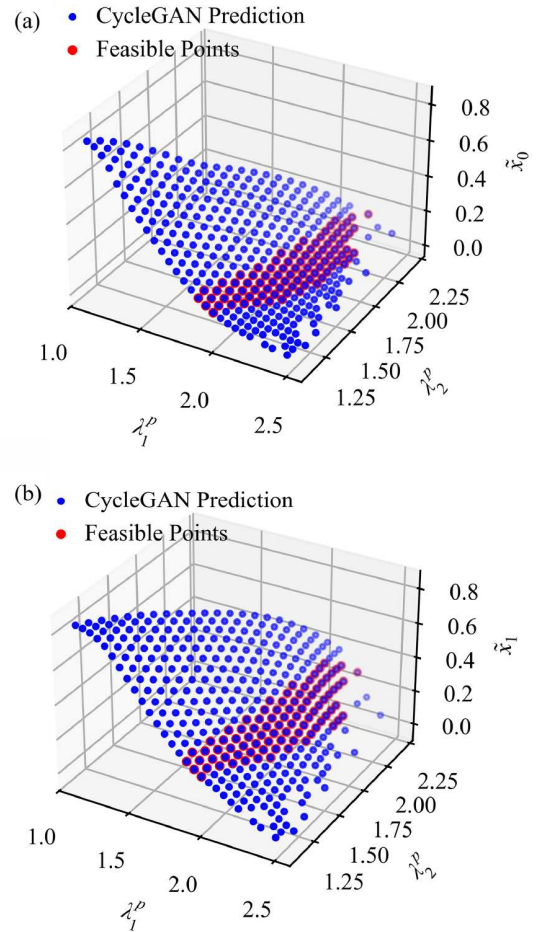


Figure 6 (Color online) Prediction results and feasible solutions of the CycleGAN model, with feasible points highlighted in red. (a) Predicted release distance; (b) predicted actuation distance.

$\lambda_2^p = 1.9$, achieves an actuation distance of 28 mm and a release distance of 13 mm, providing an operational range of 15 mm. Feasible solution 2, with prestretch ratios of $\lambda_1^p = 2.2$ and $\lambda_2^p = 1.5$, achieves an actuation distance of 18 mm and a release distance of 8 mm, resulting in an operational range of 10 mm. Additionally, in cases where specific displacement requirements—such as an operational range of 0–15 mm are not achievable, the model can also quickly predict the absence of feasible solutions. These results demonstrate the model's critical role in guiding the design and selection of DEA parameters.

Table 4 Comparison of the FEM model, MLP model, and CycleGAN model

	FEM (EXP, FEM)	MLP		CycleGAN (EXP, CycleGAN)
		(FEM, MLP)	(EXP, MLP)	
R-squared (x_0)	0.7133	0.9999	0.7361	0.9910
R-squared (x_1)	0.1204	0.9999	0.3437	0.9937
Time per prediction	ca. 8 h	ca. 0.02 s		ca. 0.73 s

5 Grasping applications

We demonstrate the DEA's application in grasping tasks. The target is to grasp objects within a 10 mm range, including thin items like paper. This requires the gripper to close after release and open to a width greater than 10 mm upon actuation. To meet these requirements, we select prestretch ratios of $\lambda_1^p = 2.5$ and $\lambda_2^p = 1.2$. The DEA-based gripper is fabricated and mounted on a Dobot magician robotic arm, which executes a gate-shaped trajectory between two points to perform pick-and-place tasks. A voltage of 6 kV is applied to open the gripper, while reducing the voltage closes it, enabling successful grasping. Using the DEA's actuation distance as a benchmark, the objects grasped were arranged by increasing size, in the following order: paper, an M2 screw, an M2 nut, an M4 screw, an M6 screw with additional weight (totaling 20 g), and an M6 nut (requiring an actuation distance of approximately 10 mm). All pick-and-place operations of the listed objects were successfully completed, as shown in Figure 7.

During the grasping process, the DEA's flexibility facilitates grasping a wide range of object sizes, providing versatility in grasping strategies. For instance, even when the size of the target object approaches the maximum actuation limit of the DEA, successful grasping can still be achieved. In the case of grasping an M6 nut, the DEA's flexibility

allows for minor collisions and interference with the object. This results in local passive deformation of the DEA, enabling it to envelop the object and complete the grasp. Similarly, when grasping thin objects such as paper, the DEA's flexibility allows the robotic arm to press the DEA down onto the table surface, leading its tip to slide against the paper. This interaction causes the paper to arch upward, facilitating a successful grasp, as shown in Figure 7(a)–(c). During the downward press, the DEA's overall configuration changes due to the displacement constraint imposed by the table surface. Once the DEA is lifted, it returns to the original configuration.

Overall, the selected DEA operates within a range of 0–10 mm and achieves a maximum grasping force of 20 g. The DEA itself weighs approximately 2.3 g, yielding a load-to-weight ratio of up to 8. This demonstrates the effectiveness of the gripper design based on the proposed approach.

6 Conclusion

In this paper, we propose a deep learning-based surrogate model for predicting the release distance and actuation distance of DEAs for various prestretch ratios. The model combines an MLP model and a CycleGAN model to achieve rapid prediction with a minimal Sim2Real gap. Specifically,

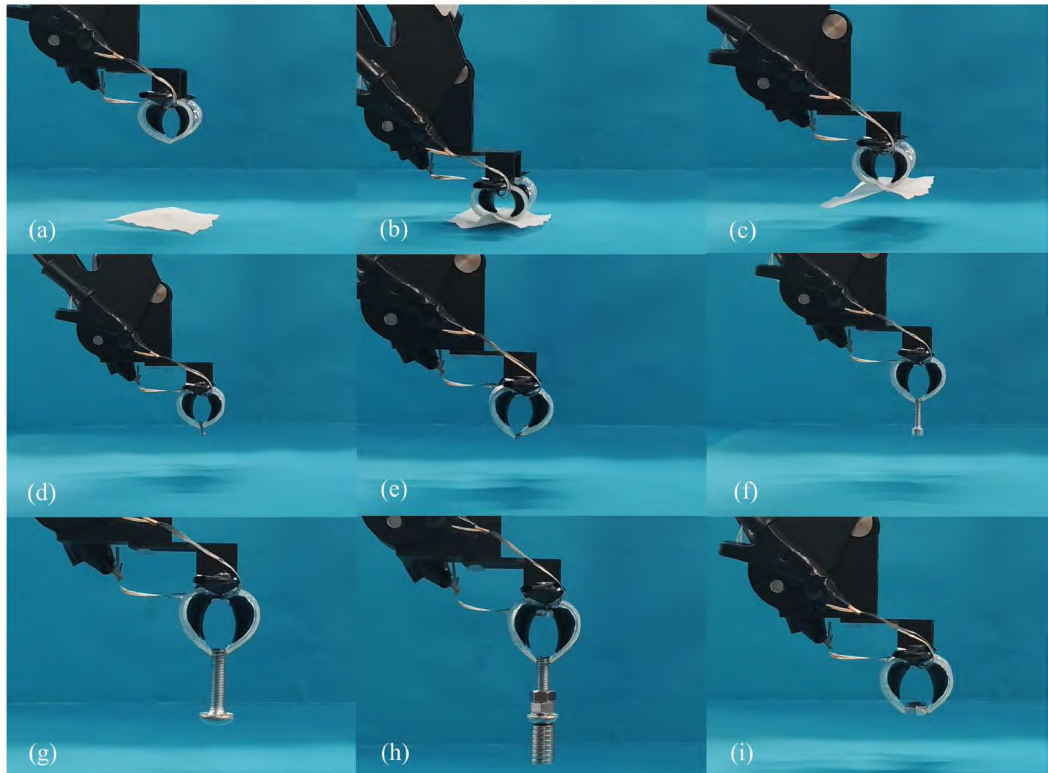


Figure 7 (Color online) Grasping experiments of objects. Grasping of tissue: (a) approaching the tissue, (b) completing the grasp, and (c) moving the tissue; grasping of screws and nuts: (d) M2 screw, (e) M2 nut, (f) M4 screw, (g) M6 screw, (h) M6 screw with additional weight (totaling 20 g), (i) M6 nut.

the MLP model significantly accelerates the prediction speed by replacing the FEM model, while the CycleGAN model improves the accuracy by training on both simulation data and experimental data. Based on the trained surrogate model, feasible design solutions can be quickly obtained for a given range of actuation strokes. Moreover, the feasible design solutions can be generalized to different scales using dimensionless parameters. The proposed approach can effectively solve the Sim2Real problem in DEA design, providing valuable guidance for practical applications. We demonstrate the DEA's application in grasping a wide variety of objects.

It should be emphasized that while the current demonstration of our proposed modeling method has been specifically conducted on an elliptical-shaped DEA, the methodology inherently possesses applicability to DEAs with diverse configurations. For novel DEA configurations, high-quality FEM-based simulation datasets remain accessible, and the Sim2Real discrepancies (material properties, manufacturing processes, assembly variations, etc.) are expected to remain stable across different configurations. Consequently, this approach sustains its practical relevance and adaptability for diverse DEA designs. In the future, we aim to extend the application of this learning-based model in DEAs of various configurations, for enhancing the performance and functionality of soft robots, such as swimming robots [3], crawling robots [4,5], and artificial muscles [39].

Acknowledgments This work was supported by the National Key Research and Development Program of China (Grant No. 2023YFB4706500), the National Natural Science Foundation of China (Grant Nos. 52275026, T2293725), the Shanghai Rising-Star Program (Grant No. 24QA2704500), and the Natural Science Foundation of Shanghai (Grant No. 23ZR1427800).

References

- Shintake J, Rosset S, Schubert B, et al. Versatile soft grippers with intrinsic electroadhesion based on multifunctional polymer actuators. *Adv Mater*, 2016, 28: 231–238
- Wang Y Z, Gupta U, Parulekar N, et al. A soft gripper of fast speed and low energy consumption. *Sci China Tech Sci*, 2019, 62: 31–38
- Li G, Chen X, Zhou F, et al. Self-powered soft robot in the Mariana Trench. *Nature*, 2021, 591: 66–71
- Chen F, Liu K, Pan Q, et al. An integrated design and fabrication strategy for planar soft dielectric elastomer actuators. *IEEE ASME Trans Mechatron*, 2020, 26: 2629–2640
- Gu G, Zou J, Zhao R, et al. Soft wall-climbing robots. *Sci Robot*, 2018, 3: eaat2874
- Gu G Y, Zhu J, Zhu L M, et al. A survey on dielectric elastomer actuators for soft robots. *Bioinspir Biomim*, 2017, 12: 011003
- Qiu Y, Zhang E, Plamthottam R, et al. Dielectric elastomer artificial muscle: Materials innovations and device explorations. *Acc Chem Res*, 2019, 52: 316–325
- Suo Z. Theory of dielectric elastomers. *Acta Mech Solid Sin*, 2010, 23: 549–578
- Chen F Y, Ren Z X, Lau G K. Maximal strengths of dielectric elastomer fingers for a passive grip. *Smart Mater Struct*, 2022, 31: 045014
- Gupta U, Qin L, Wang Y, et al. Soft robots based on dielectric elastomer actuators: A review. *Smart Mater Struct*, 2019, 28: 103002
- Chen F, Wang M Y. Design optimization of soft robots: A review of the state of the art. *IEEE Robot Automat Mag*, 2020, 27: 27–43
- Wu J, Li T, Wang J, et al. Stiffness and natural frequency of a 3-DOF parallel manipulator with consideration of additional leg candidates. *Robot Autonom Syst*, 2013, 61: 868–875
- Wu J, Gao Y, Zhang B, et al. Workspace and dynamic performance evaluation of the parallel manipulators in a spray-painting equipment. *Robot Comput-Integr Manuf*, 2017, 44: 199–207
- Wu M, Xu Y, Wang X, et al. Design and kinematics of a novel rigid-flexible coupling hybrid robot for aeroengine blades *in situ* repair. *Engin Comput*, 2024, 41: 2504–2533
- Chen K D, Zhao H, Zhang J X, et al. Modeling and control of a rigid-flexible coupling robot for narrow space manipulations. *Sci China Tech Sci*, 2025, 68: 1220302
- Luo K, Wu J N, Chen F F. Optimal biaxial prestretch ratios of soft dielectric elastomer actuators for maximal voltage-induced displacement. *Sci China Tech Sci*, 2023, 66: 2871–2881
- Chen F, Liu K, Wang Y, et al. Automatic design of soft dielectric elastomer actuators with optimal spatial electric fields. *IEEE Trans Robot*, 2019, 35: 1150–1165
- Chen B, Wang N, Zhang X, et al. Design of dielectric elastomer actuators using topology optimization on electrodes. *Smart Mater Struct*, 2020, 29: 075029
- Wang N, Chen B, Guo H, et al. Design of dielectric elastomer grippers using Bezier curves. *Mechanism Machine Theor*, 2021, 158: 104216
- Chen B, Wang N, Wang R, et al. Automatic design of dielectric elastomer-based crawling robots using shape and topology optimization. *J Mech Robotics*, 2023, 15: 021006
- Hofer S, Bekris K, Handa A, et al. Sim2Real in robotics and automation: Applications and challenges. *IEEE Trans Automat Sci Eng*, 2021, 18: 398–400
- Chen Y, Agostini L, Moretti G, et al. Dielectric elastomer materials for large-strain actuation and energy harvesting: A comparison between styrenic rubber, natural rubber and acrylic elastomer. *Smart Mater Struct*, 2019, 28: 114001
- Panahi-Sarmad M, Zahirri B, Noroozi M. Graphene-based composite for dielectric elastomer actuator: A comprehensive review. *Sens Actuat A-Phys*, 2019, 293: 222–241
- Wang H, Yang L. Effects of effective voltages, electrode types and stretching states on electrical properties and actuation characteristics of dielectric elastomer materials. *Polym Testing*, 2023, 120: 107939
- Wang H, Yang L, Zhang X, et al. Effect of different prestretching index and preloading on actuation behaviors of dielectric elastomer actuator. *J Mater Res Tech*, 2021, 15: 4064–4073
- Jayatisa S, Shim V, Anderson I A, et al. Optimization of prestretch and actuation stretch of a dea-based cell stretcher. In: Proceedings of SPIE. 2020. 11375: 148–159
- Sheng J, Chen H, Li B. Effect of temperature on the stability of dielectric elastomers. *J Phys D-Appl Phys*, 2011, 44: 365406
- Tran D Q, Li J, Xuan F, et al. Viscoelastic effects on the actuation performance of a dielectric elastomer actuator under different equal, un-equal biaxial pre-stretches. *Mater Res Express*, 2018, 5: 065303
- Kim D, Kim S H, Kim T, et al. Review of machine learning methods in soft robotics. *Plos One*, 2021, 16: e0246102
- Li Y, Liu J, Liang W, et al. Towards optimal design of dielectric elastomer actuators using a graph neural network encoder. *IEEE Robot Autom Lett*, 2023, 8: 6339–6346
- Qin Z, Ren J, Chen F, et al. Automatic design framework of dielectric elastomer actuators: Neural network-based real-time simulation, genetic algorithm-based electrode optimization, and experimental verification. *Soft Robotics*, 2024, doi: 10.1089/soro.2024.0063
- Chen W, Xu Y, Chen Z, et al. Bidirectional sim-to-real transfer for GelSight tactile sensors with CycleGAN. *IEEE Robot Autom Lett*, 2022, 7: 6187–6194
- Zhu J-Y, Park T, Isola P, et al. Unpaired image-to-image translation using cycle-consistent adversarial networks. In: Proceedings of the IEEE International Conference on Computer Vision. Venice, 2017. 2223–2232

- 34 Creswell A, White T, Dumoulin V, et al. Generative adversarial networks: An overview. [IEEE Signal Process Mag](#), 2018, 35: 53–65
- 35 Zhou Z, Lu Y, Tortós P E, et al. Addressing data imbalance in Sim2Real: ImbalSim2Real scheme and its application in finger joint stiffness self-sensing for soft robot-assisted rehabilitation. [Front Bioeng Biotechnol](#), 2024, 12: 1334643
- 36 Liao W, Wu L, Xu S, et al. A novel approach for intelligent fault diagnosis in bearing with imbalanced data based on cycle-consistent GAN. [IEEE Trans Instrum Meas](#), 2024, 73: 1–16
- 37 Araromi O A, Gavrilovich I, Shintake J, et al. Rollable multisegment dielectric elastomer minimum energy structures for a deployable microsatellite gripper. [IEEE ASME Trans Mechatron](#), 2014, 20: 438–446
- 38 Goodfellow I J, Pouget-Abadie J, Mirza M, et al. Generative adversarial nets. In: *Proceedings of Advances in Neural Information Processing Systems*. Montreal, 2014. 27
- 39 Novelli G L, Vargas G G, Andrade R M. Dielectric elastomer actuators as artificial muscles for wearable robots. [J Intelligent Material Syst Struct](#), 2023, 34: 1007–1025